

Detector Setup

BioFocus 2000

NOTE: You cannot operate in the visible range (wavelengths from 366 nm to 800 nm) without installing a tungsten lamp in your BioFocus 2000.

NOTE: For LIF detector setup, refer to “LIF Detector Setup” in this manual.

The BioFocus 2000 system employs a programmable, single wavelength UV (190-365 nm) detector. Programmable, single wavelength visible (366-800 nm) detection is available with an optional tungsten lamp.

When you click **Detector**, a Detector Setup screen (see Figure 5-37) appears, showing several options.

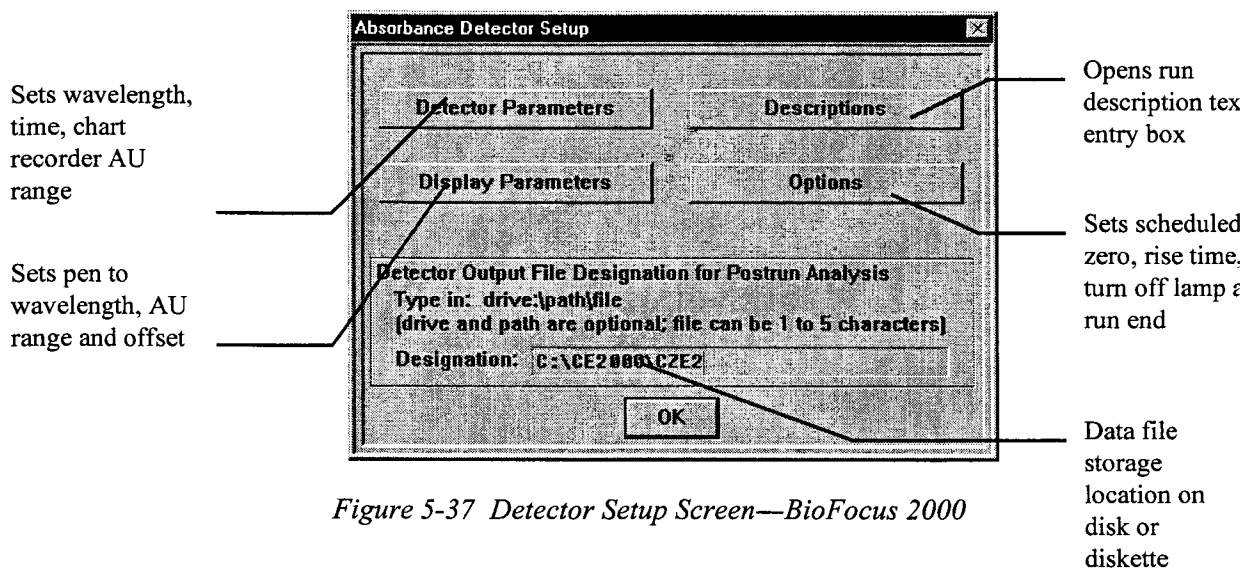


Figure 5-37 Detector Setup Screen—BioFocus 2000

Detector Parameters

Clicking **Detector Parameters** calls up the Single Wavelength-Time Programmable screen shown in Figure 5-38. You can enter parameters such as detection wavelength, total run time, strip chart channel AU Range, and Autozero using this screen.

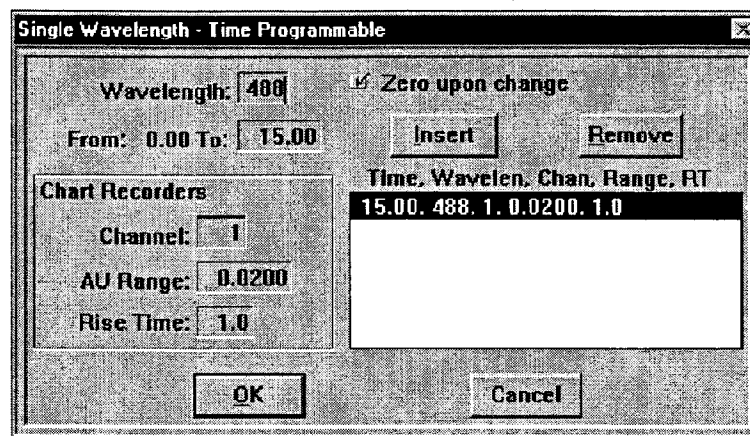


Figure 5-38 Single Wavelength-Time Programmable Screen—BioFocus 2000

You can either process a run at a single wavelength for the entire run time, or you can process a run at different wavelengths at different times during the run (up to eight wavelengths may be extended). Run times must be programmed for a minimum of 1 minute to avoid display problems (run times can also be entered directly from the Define Method screen by typing in a number). The software will not accept run times of less than 1 minute or greater than 120 minutes.

Adding and Deleting Wavelengths

Use the **Insert** and **Remove** commands to add and delete wavelengths. Type in the wavelength and a stop time for each wavelength in the appropriate boxes. Select the desired strip chart channel absorbance range from a preset list of choices (from 0.0005 to 2.000, spaced at discrete intervals). If you are programming a single wavelength for the entire run, the stop time equals the total run time. If you program several wavelength changes during a run using the From: To: box, you have the option of automatically zeroing the baseline after each wavelength change by clicking **Zero upon change**. The stop time for the final wavelength change defines the total run time. Please note that integration cannot be performed when wavelength changes have been programmed. Click **OK** when you are finished with this screen.

NOTE: Whenever you change a wavelength from the preset value under Detector parameters, you will need to change the Display Parameters since the default display changes to “none” if a wavelength other than the preset is programmed

Choosing Display Parameters

Display Parameters, shown in Figure 5-39, allows you to assign pens for up to three traces. You can choose values for each pen color: wavelength, absorbance units, and baseline AU offset (-10 to 75% of full scale). Always click **Select Pen** first. Click **OK** after you have made your selections.

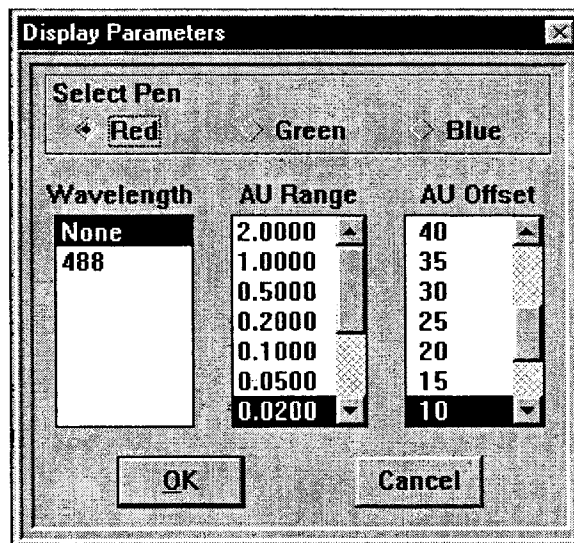


Figure 5-39 Display Parameters Screen—BioFocus 2000

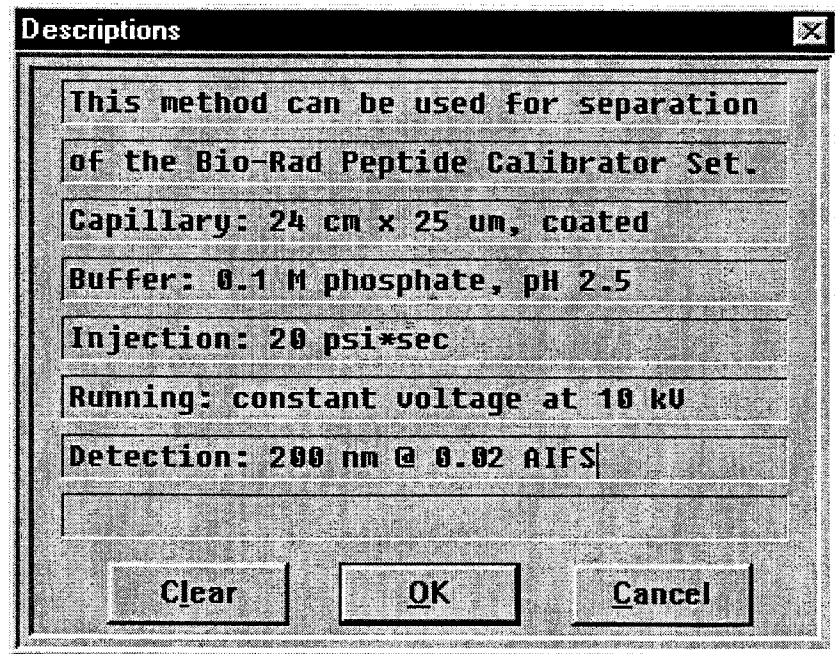
Descriptions and Options

Use of these items on the main Detector Setup screen is optional:

The Descriptions box is provided for your documentation purposes only.

The default Descriptions box (see Figure 5-40) for the template methods is blank. When using template methods for programming a new method, the descriptions for the new method can be entered in this box.

NOTE: When using existing methods as templates for programming a new method, the text in the Descriptions box should be edited as appropriate for the new method, or be deleted. If not, it will be added to the raw data files generated when the new method is used, and will be displayed in the View Electropherogram post-run raw data file index.



Descriptions

This method can be used for separation
of the Bio-Rad Peptide Calibrator Set.

Capillary: 24 cm x 25 um, coated

Buffer: 0.1 M phosphate, pH 2.5

Injection: 20 psi*sec

Running: constant voltage at 10 kV

Detection: 200 nm @ 0.02 AIFS

Clear OK Cancel

Figure 5-40 Descriptions Box

If you wish to delete the text in the Descriptions box, click the **Clear** button.

NOTE: The actual analysis parameters are automatically incorporated into the raw data file by the BioFocus software, and this information is automatically printed below the electropherogram.

The Options screen, shown in Figure 5-41, allows you to program an additional time point for re-zeroing the baseline during the analysis. Scheduled zero is entered as time in minutes. You can also select a rise time from a preset list of choices, ranging from 0 to 5 seconds. There is also a lamp turn-off feature that can be used at the end of an autosequence to extend lamp life.

NOTE: Use of the lamp turn-off feature produces baseline drift during the next run if lamp warm-up time is too short.

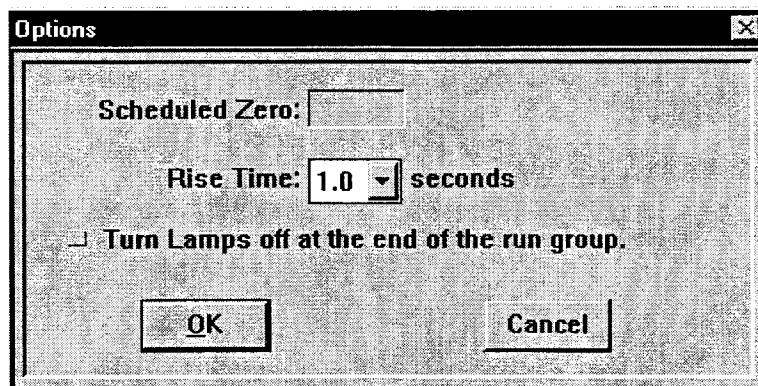


Figure 5-41 Options Screen—BioFocus 2000

Saving Data Files for Later Analysis

Enter an optional drive:path and a required file name of 1-5 characters in the Designation box. The file pictured in Figure 5-42 is called “CZE2”. All detector files will have a BFF extension (suffix) and be numbered in the order they are collected for each method. If the optional drive:path is not designated, the raw data files are stored on the hard disk in the directory that contains the main application program, usually but not necessarily CE2000. For convenience in identifying these files after analysis (for displaying or printing electropherograms, or current, voltage, and temperature histories), it is recommended that each method be assigned a unique file name. The file name appears above the electropherogram on the computer screen during the actual data collection phase of the run.

NOTE: Be sure to name your Detector Output Files for identification during post-run analysis

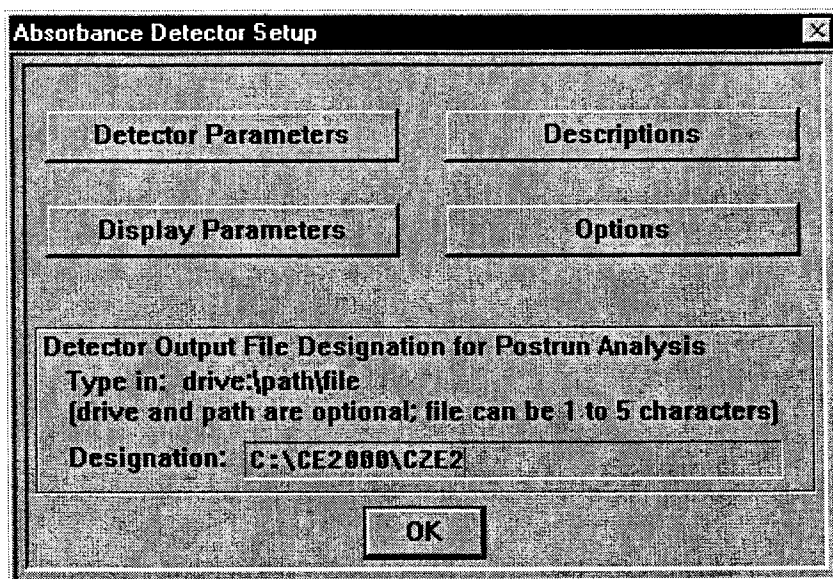


Figure 5-42 Detector Setup Screen—BioFocus 2000

NOTE: The newly-specified detector parameters now appear next to the Detector button.

Click **OK** at the bottom of the Detector Setup screen when you are finished defining all detector functions. This returns you to the Define Method screen.

BioFocus 3000

NOTE: For LIF detector setup, refer to “Defining LIF Methods” in this section.

BioFocus 3000 detector setup is very similar to BioFocus 2000 setup, but you have the additional option of choosing multiple wavelength detection and high speed scanning. The follow procedures describe only those detector features unique to the BioFocus 3000. Refer to the BioFocus 2000 detector setup procedures in the preceding paragraphs for setup procedures common to both systems.

When you click **Detector**, the BioFocus 3000 Detector Setup screen appears. Figure 5-43 shows an example of a Setup screen.

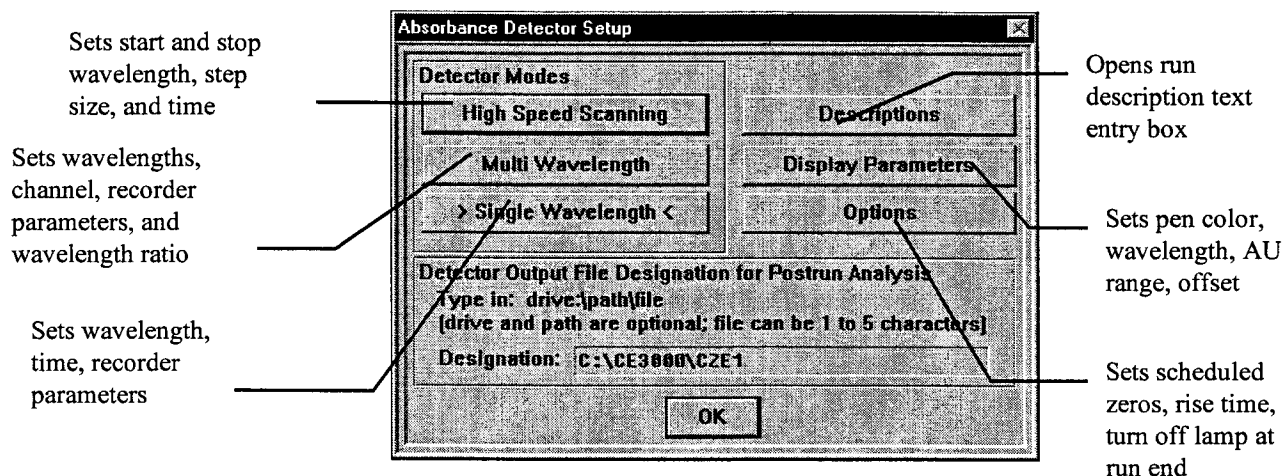


Figure 5-43 Detector Setup Screen—BioFocus 3000

Note: The software will not accept run times of less than 1 minute or greater than 120 minutes.

Detector Overview

You can choose High Speed Scanning mode (acquisition of spectra over a defined wavelength range), Multi Wavelength mode (simultaneous monitoring of a maximum of 32 wavelengths), or Single Wavelength mode for your method. Click the desired mode. The detector function you choose will be labeled by angle brackets (> and <). Run times for any mode must be programmed for a minimum of 1 minute to avoid display problems (run times can also be entered directly from the Define Method screen by typing in a number).

Setting Up a Detector Mode

Use this procedure to set up a detector mode for the method that you're defining:

NOTE: It is not possible to scan both UV and visible wavelengths in the same run. The scan rate will be determined by the scan range, the step size, and the collection rate of 96 points per second. For example, scanning a range of 200-320 nm at a step size of 5 nm will produce a scan rate of approximately 4 Hz.

1. Choose a mode by clicking High Speed Scanning, Multi Wavelength, or Single Wavelength.
2. If you choose High Speed Scanning, you will see another screen (Figure 5-44) where you enter values for the following: the wavelength range you wish to scan between 190 and 365 nm or between 366 and 799 nm (start and stop), the interval between wavelengths (step size), and the total run time (stop time). Click **OK** to save the information and return to the detector setup screen.

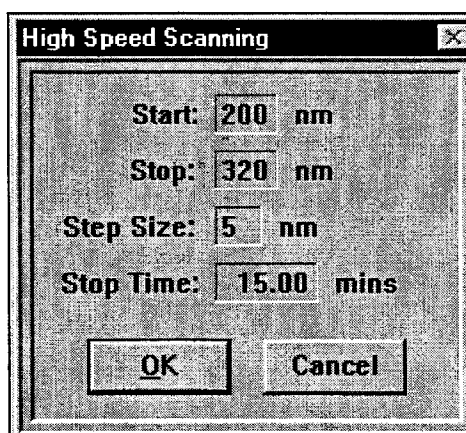


Figure 5-44 High Speed Scanning Screen—BioFocus 3000

3. If you choose Multi Wavelength, you will see the screen shown in Figure 5-45. Enter the wavelengths that you wish to use when running your method. Also, enter strip chart parameters. If you wish to have the software calculate ratios of absorbance for two wavelengths, enter ratio information. Click **OK** to save the information and return to the detector setup screen.
4. If you select Single Wavelength, the screen shown in Figure 5-46 appears. Enter a wavelength and a time that you wish for the wavelength to be monitored. You can enter more than one wavelength, but the total time must be equal to the run time. Enter chart recorder information if required. Click **OK** to save information and return to the detector setup screen.
5. If you wish to document the method, click **Descriptions**. This brings up the screen shown in Figure 5-40. Enter any pertinent data. Click **OK** to save the information and return to the detector setup screen.
6. Click **Display Parameters** to bring up the screen shown in Figure 5-39. Select a wavelength, a pen color, and AU range and offset.

Click **OK** to save the information and return to the detector setup screen.

7. Click **Options** to see the screen shown in Figure 5-41. Enter times for scheduled zeros, a rise time number, and whether you wish to turn off lamps at the end of a run. Click **OK** to save the information and return to the detector setup screen.
8. Enter a path for your output files in the **Designation** box.

This completes detector setup. Click **OK** to return to the **Define Method** screen.

High Speed Scanning Considerations

The maximum wavelength should be an exact multiple of the step size from the minimum wavelength. For example, if you wish to scan at 10 nm intervals throughout the UV range, starting at 195 nm, you should enter either 365 or 355 nm as your maximum wavelength, but not 360 nm. If you do not specify a scan range where the higher value is an exact multiple of the step size starting from the lower value, the program will truncate to the nearest multiple. For example, if you specify a range from 200-365 nm at 10 nm step sizes, the last wavelength that will be scanned is 360 nm (the **Display** window during an automation sequence can be checked to verify which wavelengths are being monitored).

NOTE: There is no external integrator signal output during high-speed scanning.

Maximum Number of Wavelengths

Only three wavelengths can be monitored at one time on the computer screen. However, all wavelengths (user-defined interval values) can be displayed simultaneously in the post-run **Spectra** program, and individual wavelength electropherograms may be analyzed in the post-run **Bio-Rad Integrator** program.

Spectra Display

Spectra can be displayed following the completion of an automation sequence with the **Spectra** software program. The **Spectra** software program can be accessed from the **Main Screen Toolbar**.

Multi Wavelength

Clicking **Multi Wavelength** allows you to monitor up to 32 different detection wavelengths, either in the range 190 nm through 365 nm or in the range 366 nm through 800 nm. Up to three wavelengths can be displayed on the **BioFocus 3000** screen during an automation sequence (see Figure 5-45). Three separate chart recorder channels are available for any three wavelengths plus a fourth channel for the ratio of any two of the three wavelengths.



Spectra Button

NOTE: Detection requires significant amounts of target disk space for data storage. It is recommended that these data files be frequently transferred to a floppy or external disk for storage, then deleted from the target disk when routinely operating the detector in this mode.

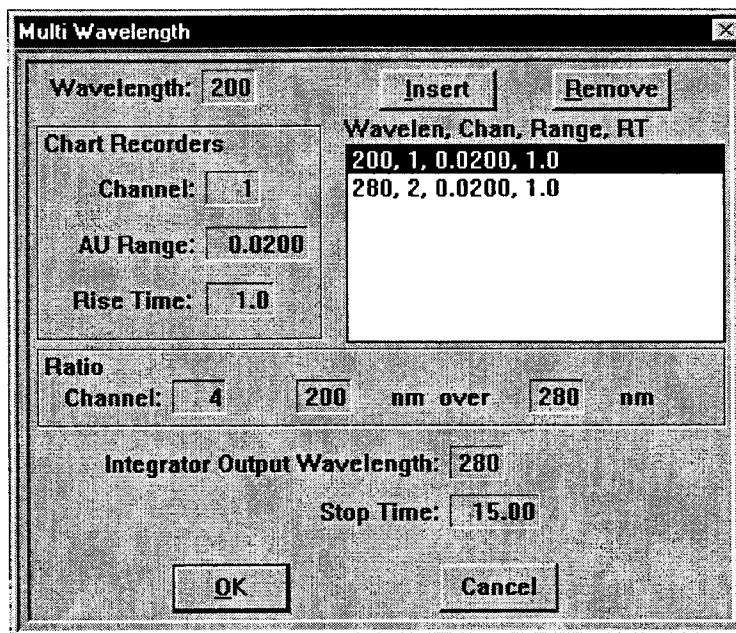


Figure 5-45 Multi-Wavelength Screen—BioFocus 3000

NOTE: Before attempting to program wavelengths in the visible region (366-800 nm) in the Multi Wavelength screen, all UV wavelengths in the Ratio Boxes and wavelength table must be deleted except the first one. Using the **Remove** button, delete all but the first line. Then enter the desired wavelength in the Wavelength box. Additional detector wavelengths in the visible region may be added as additional lines in the table.

Inserting and Removing Wavelengths

Use the **Insert** and **Remove** commands to add and delete wavelengths in the box on the right of the screen. Figure 5-46 shows an example. To insert a new wavelength, first click insert, then type in the wavelength, then click another box. The wavelength now appears in the box on the right. Type in the strip chart channel, absorbance units, and rise time in the appropriate boxes on the left in the same way. Type in a stop time—this is the same as total run time. Make sure the wavelengths in the Ratio Channel correspond to wavelengths listed in the upper Wavelength box. Click **OK** when you are finished.

The screenshot shows the 'Multi Wavelength' dialog box with the following fields and table:

- Wavelength:** 214
- Insert** and **Remove** buttons
- Chart Recorders** section:
 - Channel:** 3
 - AU Range:** 0.0500
 - Rise Time:** 1.0
- Ratio** section:
 - Channel:** 4
 - 214 nm over 280 nm**
- Integrator Output Wavelength:** 280
- Stop Time:** 15.00
- OK** and **Cancel** buttons
- Wavelength, Chan, Range, RT** table:

Wavelength	Chan	Range	RT
200	1	0.0200	1.0
214	3	0.0500	1.0
280	2	0.0200	1.0

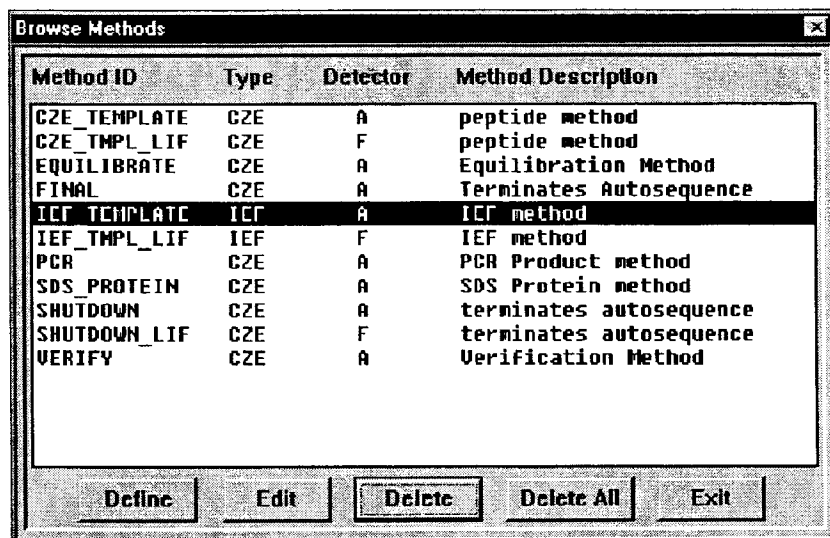
Figure 5-46 Multi-Wavelength Screen—With Wavelength Inserted

LIF² Detector Setup

NOTE: Before beginning method definition, review filter and emission wavelength information. From the main screen, click **Commands** and then **LIF Detector Setup**. If you wish to change filter and emission wavelength information, use the Detector Setup screen. You cannot edit this information directly once you start method definition. You may, however, define and save a method, change filter and wavelength information, and then edit the method.

The process of defining LIF methods is very similar to the process of defining UV/Vis (absorbance) methods, described in “Defining Absorbance Methods” on page 29. The following paragraphs describe the differences between defining LIF and UV/Vis methods. The major difference lies in detector setup.

Figure 5-47 shows the Browse Methods screen that appears when you click the Methods button on the main screen. Note that some method templates have an “F” in the Detector column. These are LIF methods; absorbance methods have an “A” in the Detector column. You cannot define an absorbance method linked to an absorbance automation sequence if you are operating in LIF mode.



Method ID	Type	Detector	Method Description
CZE_TEMPLATE	CZE	A	peptide method
CZE_TMPL_LIF	CZE	F	peptide method
EQUILIBRATE	CZE	A	Equilibration Method
FINAL	CZE	A	Terminates Autosequence
IEF_TEMPLATE	IEF	A	IEF method
IEF_TMPL_LIF	IEF	F	IEF method
PCR	CZE	A	PCR Product method
SDS_PROTEIN	CZE	A	SDS Protein method
SHUTDOWN	CZE	A	terminates autosequence
SHUTDOWN_LIF	CZE	F	terminates autosequence
VERIFY	CZE	A	Verification Method

Define Edit Delete Delete All Exit

Figure 5-47 Browse Methods Screen

LIF Method Setup Example

Follow the procedure given in this example to set up an LIF method using an LIF method template.

Select CZE_TMPL_LIF for definition by double-clicking it in the Browse Methods listing. Figure 5-48 shows the screen that appears. Note that some parameters have default values. Also note that there is an indication opposite the **Detector** button that lists two internal lasers and their wavelengths, together with gain settings for two PMTs.

The 'Define Method' dialog box is shown with the following fields and controls:

- Method ID:** CZE_TMPL_LIF
- Description:** peptide method
- Buttons:** OK, Cancel, Inlet Reagent, Outlet Reagent, PO4_RUN
- Run Parameters:**
 - Voltage Mode:** Run Voltage: 10.0 kV, Polarity: + + + - > - + +
 - Current Mode:** Current Limit: 50.0 μ A, Capillary Temperature: 20 $^{\circ}$ C, Run Time: 12.00 minute(s)
- Preparation Cycle Parameters:** 1 cycle(s) defined

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Pre-inj	PO4_PREP	Waste	High	60	sec
- Buttons:** Add, Insert, Delete, Delete All
- Injection:** Pressure, 20 (psi * sec)
- Detector:** Laser 488 nm Laser 594 nm PMT1 1000 X Gain PMT2 1000 X Gain
- Timed Events:**

Figure 5-48 LIF Method Definition Screen

Click the **Detector** button to bring up the LIF Detector Setup screen shown in Figure 5-49.

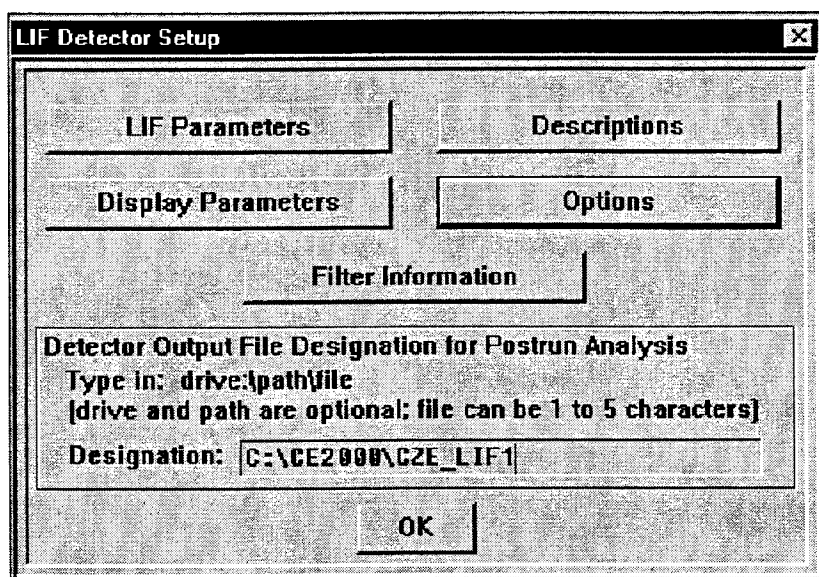


Figure 5-49 LIF Detector Setup Screen

Setting Up Laser and PMT Detector Parameters

You can set laser and PMT detector parameters for the method from this screen. You can also record comments that the software retains with data for a run using this method. Before you exit this screen after completing data entry, ensure that you have specified a path and file name for the detector output file saved after a run.

Click the **LIF Parameters** button to bring up the screen shown in Figure 5-52. Although there are default parameters set for this method, you can change PMT gain settings and turn lasers on or off according to your requirements. The screen shown here is for a system with two internal lasers and two PMT detectors. The screen that appears when you define this method reflects your system configuration, as determined by configuration switch settings made on the LIF Module. Click **OK** to exit the screen and return to the method definition screen. If you did not turn on at least one of the lasers or you set both PMT gains to zero in the LIF Parameters screen, the error message shown in Figure 5-53 appears to remind you to make correct settings.

Selecting Display Parameters

To select LIF display parameters, click **Display Parameters** from the LIF Detector Setup screen. The screen shown in Figure 5-50 appears.

The dialog box is titled "LIF Display Parameters" and contains settings for two channels. Channel 1 settings are as follows:

CHANNEL 1	
1,000	30
500	25
200	20
100	15
50	10
20	5
10	0

Range (RFU) Offset (%)

Pen Color

- > Red
- > Yellow
- ◀ Green
- > Blue

Background Color

- ◀ Black
- > White

Channel 2 settings are identical to Channel 1.

Buttons: OK, Cancel

Figure 5-50 LIF Display Parameters Screen for First PMT Channel

Description and Options

Use of these items on the LIF Detector Setup screen is optional.

The Descriptions box is provided for your documentation purposes only.

The default Descriptions box (see Figure 5-51) for the detector setup is blank. Use this screen to enter information about setup or any other remarks you wish to make.

If you wish to delete the text in the Descriptions box, click the **Clear** button.

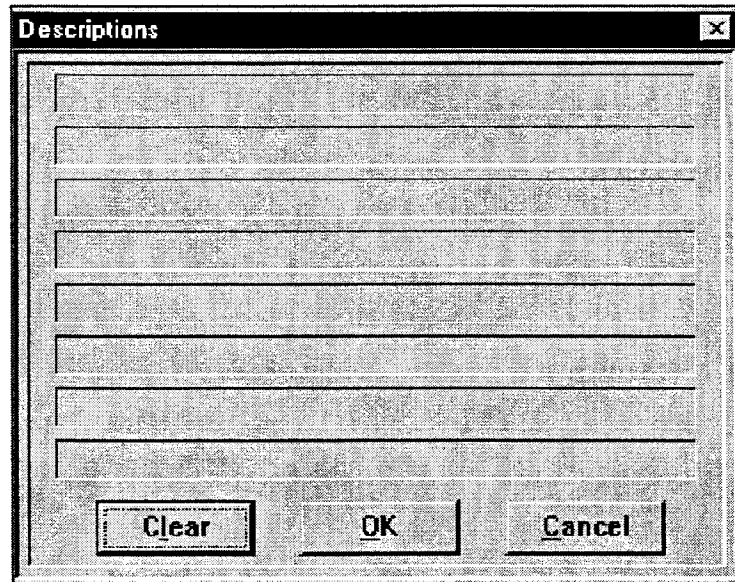


Figure 5-51 Description Screen

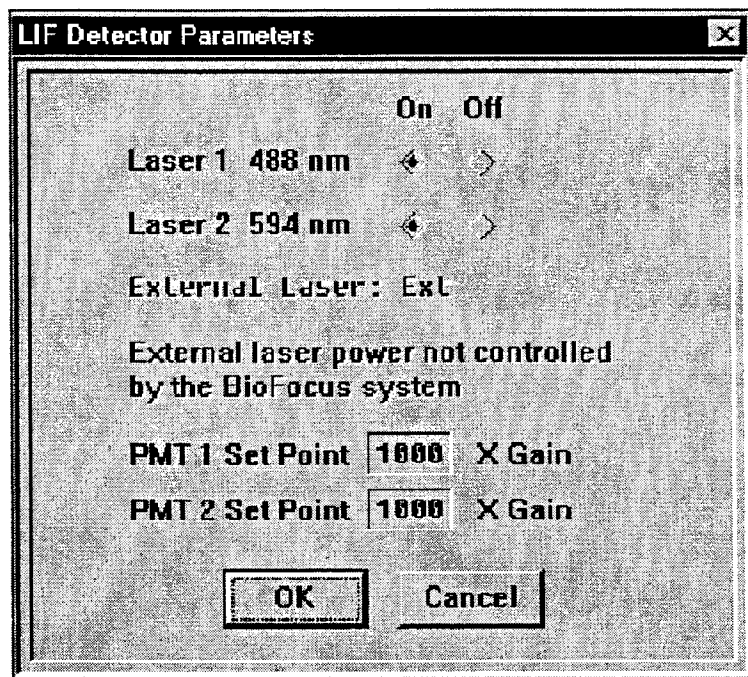


Figure 5-52 LIF Parameters Screen

Selecting Options

Click the **Options** button to move to the Options screen shown in Figure 5-55. Enter a rise time for the run (valid entries are from 0.0 to 10 seconds) and determine whether you want the laser(s) turned off at the end of the run. Click **OK** to exit the screen and return to the method definition screen.

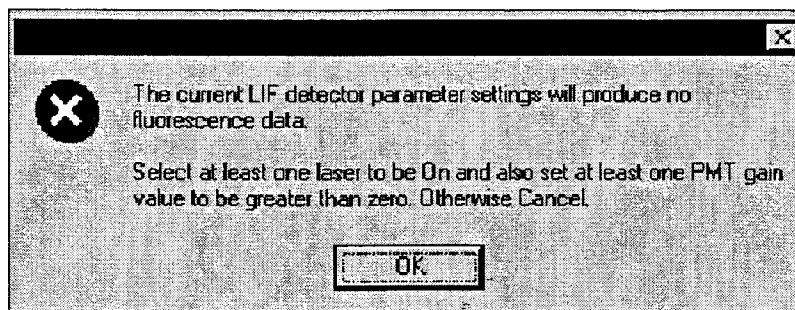


Figure 5-53 Detector Setup Error Message

Filter Information

Click the **Filter Information** button to bring up the screen shown in Figure 5-54. You can review the filter information shown, but you cannot edit it. To edit filter information, return to the main screen, click **Commands** and then **LIF Detector Setup**. A screen appears to allow you to enter filter and emission wavelength information. Click **OK** to exit the screen and move to the method definition screen by clicking the **Methods** button and choosing a template.

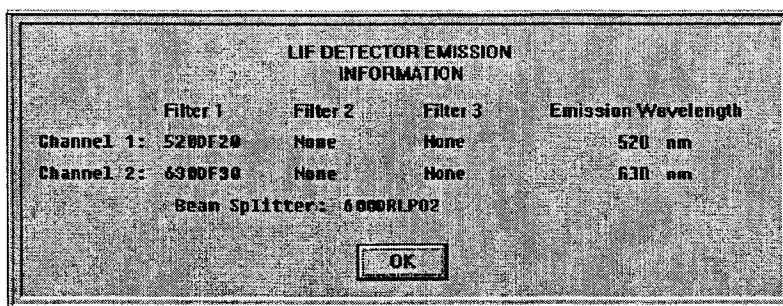


Figure 5-54 Filter Information Screen

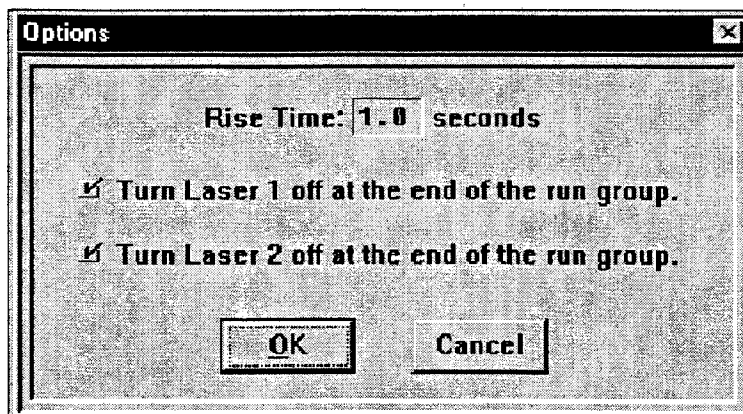


Figure 5-55 Options Screen

From this screen, set the parameters for electropherogram display, including background color, pen color, range, and offset. Note that in a single-channel system, channel 2 information will be grayed out.

When you complete detector setup, click **OK** to return to the method definition screen. Complete method definition as outlined in "Defining Absorbance Methods" on page 29.

Gradient Methods (Timed Events)

A gradient method uses timed events to cause the BioFocus system to change capillary voltage and temperature in a linear fashion between programmed event times during a run. The user programs event times on the Timed Events screen, reached from one of the Define Method screens. Programming events also allows the user to set the pressure mode at the selected event times to Off, Low (approximately 5 psi), or High (approximately 100 psi).

A CZE method set to Voltage mode or an IEF method (which is unconditionally Voltage Mode) provide the full capability of specifying voltage, capillary temperature, and pressure mode at the desired times. When set to Current mode, a CZE method is limited to specifying capillary temperature and pressure mode at the programmed times.

Main Method Screen Access To Timed Events

To access the Timed Events screen, move to the main BioFocus screen. Click **Methods** to move to the Browse Methods screen. Double-click the template in which you wish to use to define Timed Events: either CZE method, the IEF method, the CZE single run group, or the IEF single run group. From the Define Method screen, click **Timed Events** to bring up the Timed Events screen shown in Figure 5-56.

Time (min)	Voltage (kV)	Temperature (°C)	Pressure (mode)
1.0	5.0	15	Low
1.9	10.0	25	Low
2.0	7.5	15	Low
3.0	7.5	15	Low

Buttons: Add, Edit, Insert, Delete, Delete All, OK, Cancel

Figure 5-56 Timed Events Screen

You can use the mouse to select a field on the Add Timed Events screen, or use the **Add** button to add a new event. When you complete parameter specification for the first timed event, click **OK** to return to the

Timed Events screen. You can enter additional timed events by using the procedure described in the paragraph above.

Main Method Screen Timed Event Associations

The main method screens set parameters for methods without timed events. Note that both the run (or mobilizing) voltage and the capillary temperature fields display the associated setpoints and are editable. With timed events set, the voltage and capillary temperature fields are not editable and display the word “Prog”. Since these are timed events, they are accessible on the Timed Events screen. The Current field displays the associated setpoint and is editable, while the Voltage Mode Run Voltage field is not editable and displays the word “Prog”.

NOTE: In general, pressing a down arrow key highlights the following row while pressing an up arrow key highlights the previous row. Pressing a right or left arrow key can produce a spiraling effect: if the cell highlight is in the first or last column of any row, pressing a right arrow key moves the highlight to the first column of the following row; if the cell highlight is on the first column of any row, pressing a left arrow key moves the highlight to the last column of the previous row.

Timed Events User Interface

The Timed Events screen is in a spreadsheet format with four columns (time, voltage, temperature, and pressure) and (via scrolling) 50 rows (timed events). Zero time is the start of electrophoretic separation: either the start of the CZE run phase or the start of the IEF mobilizing phase. Time separation between adjacent timed events must be equal to or greater than 0.1 minute.

The Timed Events screen itself is effectively read-only. You can edit the entries by using the buttons directly below the screen, or by using the scroll bar, specific keyboard keys, or by clicking or double-clicking within the data portion of the screen. Additional screens (described shortly) provide you with editing capabilities.

Adding a Timed Event

To access the Add Timed Events screen, either click the **Add** button, press one of the arrow keys, or move the mouse cursor to any cell of an unused row and click the left mouse button. Adding a timed event places a new event after the last event in the table. Adding is possible only if the current final time is less than 120.0 minutes and also only if the current number of events is less than 50. If adding is not possible when you try to add an event, a suitable message will advise you why adding is not possible.

NOTE: If the cell highlight is in the last column, you can move to the first column of the row to be added by pressing the right arrow key.

To add, click the Add button, click the scroll bar's down arrow, or click anywhere in the spreadsheet below the current final event. If the current number of events is 15 or fewer, you may also click in the scroll bar below the scroll bar slider. Finally, if the current final event is highlighted, you may add an event by pressing a down arrow key. Refer to Figure 5-57.

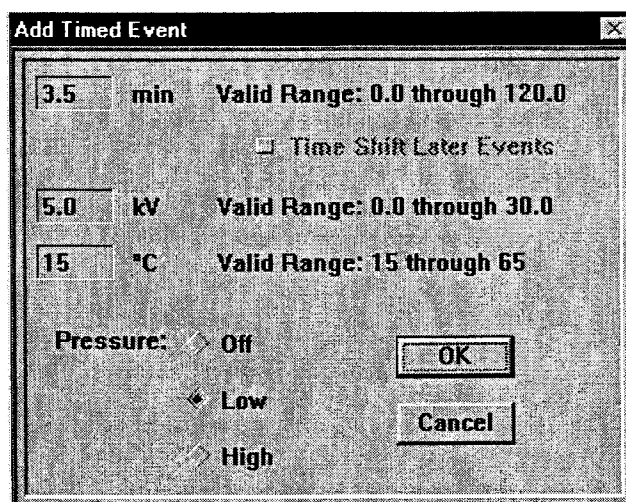


Figure 5-57 Add Timed Event Screen

The Add Timed Event screen provides you with the means to modify voltage and temperature parameters at scheduled times. The default parameters are the previous final event's time plus 0.1 minute. The remaining parameters are the previous final event's corresponding parameters.

Each editable field is identified by units with a valid range beside it. Also, there are three buttons from which you may select the pressure mode. Note that the Time Shift Later Events check box is grayed out. This feature is active only for editing or inserting a Timed Event, and will be described later.

If the parameters are within their valid ranges, the **OK** button saves them and returns you to the main Timed Events screen. Otherwise, a message advises you that a parameter is not within its valid range and keeps you on the Add Timed Event screen. The **Cancel** button returns you to the main Timed Events screen without saving the parameters you set.

Editing a Timed Event

To edit the highlighted timed event, click the **Edit** button or to edit any timed event, double-click its row. If there are no defined events, Editing reverts to Adding. Editing an event cannot change its relative pre-edit position with respect to the other events.

The Edit Timed Event screen is similar to the Add Timed Event screen with the addition of the activation of the Time Shift Later Events check box. See Figure 5-58. The default parameters are those that currently exist for the event.

Time shifting will apply equally and only to each timed event following the event being edited. In other words, time shifting moves all later events as a single block in time.

The **OK** and **Cancel** buttons have the same function as their counterparts on the Add Timed Event screen.

Edit Timed Event

5.5 min Valid Range: 5.1 through 120.0

☐ Time Shift Later Events

7.5 kV Valid Range: 0.0 through 30.0

15 °C Valid Range: 15 through 65

Pressure: ☒ Off ☐ Low ☐ High

OK Cancel

Figure 5-58 Edit Timed Event Screen

Inserting a Timed Event

Inserting is possible only if the current number of events is fewer than 50 and also only if there is an available time for the insertion, based on the requirement (stated above) that the time separation between adjacent events must be equal to or greater than 0.1 minute. If there are no defined events, Inserting reverts to Adding. Inserting an event cannot change the pre-insert relative positions of events before and after the insertion point. If inserting is not possible, an error message will advise you why inserting is not possible.

To insert an event immediately above the highlighted event, click the **Insert** button. A screen very similar to that shown in Figure 5-58 appears.

The Insert Timed Event screen is the same as the Edit Timed Event screen, with a different screen title. The default parameters are the pre-insert time of the event immediately before the inserted event +0.1 minute, time ramping indication (– sign) (the following section will discuss time ramping) for voltage and cartridge temperature, and the pre-insert pressure mode of the event immediately before the inserted event. You must delete the – sign in a field before you can enter a valid value.

Time shifting and the **OK** or **Cancel** buttons have the same function as their counterparts on the Edit Timed Event screen.

Time Ramping

Time ramping occurs automatically for voltage and cartridge temperature event entries that display the time ramping indication (–sign). The user may assign time ramping to any event's voltage and/or cartridge temperature entries, through the Add Timed Event, Edit Timed Event, and Insert Timed Event screens.

The value of each entry displaying a – sign is the linearly interpolated (ramped) value with respect to time, from the nearest preceding event entry to the nearest succeeding event entry.

When the user clicks the **OK** button, the software will not accept zero time event entries or final time event entries that are time ramping entries.

The maximum rate of change of temperature is 80° C per minute. If you enter a value that exceeds this rate, a message appears informing you of the error. You cannot proceed to the next time entry until you correct the error.

Deleting a Timed Event

To delete the highlighted event, click the **Delete** button. After you confirm the deletion, the event clears from the screen.

Deleting All Timed Events

To delete all events, click the **Delete All** button. After you confirm removing all events, all events clear from the screen.

Saving Timed Events

To save the events you programmed, click the **OK** button. This saves the list of events and then returns you to the main method screen.

Canceling Modifications to the Timed Events Table

To cancel modifications to the event table, click the **Cancel** button. This will restore the event table to its state before you entered the Timed Event screen and then return you to the main method screen.

Timed Events Vs Method Run Time

In a CZE method, the zero time begins with the electrophoretic separation phase of a run. In an IEF separation method, zero time begins earlier. Method run time begins with the storing of detector data in a run's *.BFF data file.

For CZE methods, zero time for timed events corresponds to the start of method run time. For IEF methods, however, run time begins with focusing, while zero time for timed events begins with mobilization. Thus, during an IEF run, a particular timed event will occur at a run time which is equal to the time for focusing plus the time of the event, as shown on the Timed Events screen.

After you have specified all parameters, be sure to click **OK** in the upper right corner of the Define Method screen to save everything you have created for this method. Now you are back in the Browse Methods Screen shown in Figure 5-59.

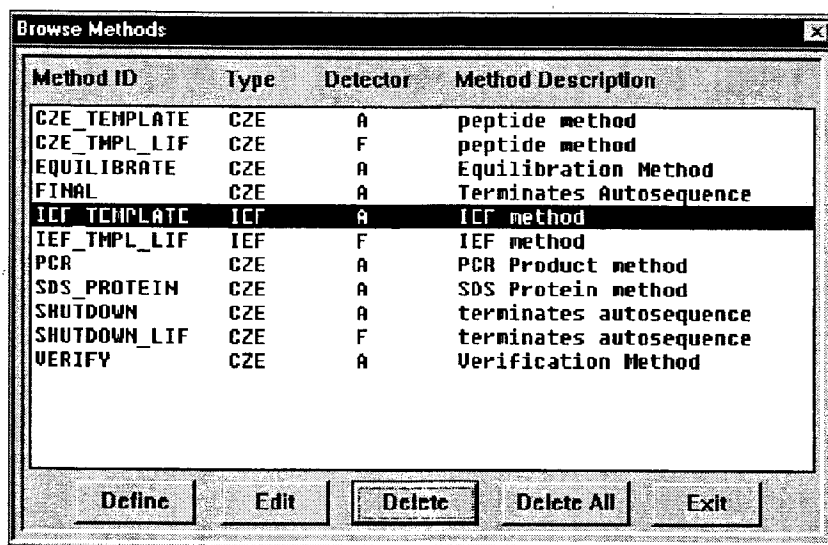


Figure 5-59 Browse Methods Screen

Editing Methods

You can edit any of the methods by clicking the **Edit** button, except any template method or any user-programmed method already linked to an automation sequence.

Deleting Methods

You can delete a method (except for linked or reference methods) by highlighting the method and clicking the **Delete** button.

Deleting All Methods

All methods, with the exception of linked or reference methods, can be deleted by clicking the **Delete All** button. A dialog box appears to allow you to continue the deletion or to cancel.

Clicking the **Exit** button brings you back to the Main Screen.



Print Requests Button

If you wish to make a hard copy of the new method, you can do so from the Print Requests button on the toolbar (this button becomes inactive during the run).

The CZE template may also be used for defining other methods, including Micellar Electrokinetic Chromatography (MEKC), Capillary Ion Analysis (CIA), Capillary Gel Electrophoresis (CGE), and Chiral Analysis.

IEF Method

NOTE: Injection in IEF mode is done under high pressure (100 psi) to ensure that the capillary is completely filled with a homogeneous solution.

If you double-click the IEF template in the Browse Methods screen, a Define Method screen appears. See Figure 5-60. Any of the predefined run parameters displayed on this screen can now be changed to define a new IEF method. You should enter a method ID and a description in the appropriate boxes. In the Inject Time you indicate the capillary filling time for the sample plus ampholyte mixture. For a 24 cm x 50 μ m capillary, the recommended sample injection time is 20 seconds. On this screen you also choose Polarity (always + $\rightarrow$ - with standard Bio-Rad CIEF reagents) and Cartridge Temperature. The recommended cartridge and carousel temperatures for IEF are 27° C and 20° C, respectively.

Define Method

Method ID: IEF_TEMPLATE OK Cancel

Description: IEF method Polarity: + - - > - + +

Inject Time: 20 sec Run Time: 30.00 min Capillary Temperature: 27 °C

Detector: Single Wavelength Mode, 280 nm

Timed Events

Preparation Cycle Parameters: 0 cycle[s] defined

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
Add Insert Delete Delete All						

Step	Inlet	Outlet	Terminate Focus by	End Value	Voltage (kV)	Limit (µA)
Focus	IEFANOLYTE	IEFCATHOLYTE	Time(sec)	240	15.0	100.0
Mobilize	IEFANOLYTE	IEFMobilizer			15.0	100.0

Figure 5-60 Define Method Screen—IEF

NOTE: Run time equals focusing time plus mobilization time.

Detector Options

If you click **Detector**, you see the same options described in the CZE method template for wavelength, run time, sensitivity, display parameters, and naming the detector output files. The wavelength usually employed in IEF is 280 nm because ampholytes absorb at 200 nm.

Timed Events

You can create methods that contain timed events. These events can be used to change run voltage or cartridge temperature at predefined times during a run. For a complete description of timed events, refer to “Gradient Methods (Timed Events)” earlier in this section.

NOTE: The Bio-Rad supplied IEF template comes without prep cycles defined.

Prep Cycle Parameters

Prep cycle parameters can be defined from the Prep Cycle Parameters Table. See “Entering Preparation Cycle Parameters” on page 5-34.

Multi Inject Parameters

If you wish to use Multi Inject with CIEF, you should inject the internal standards (prepared in ampholytes) after the sample using a low-pressure Post Inject cycle. For this cycle, select standards as the Inlet and Waste as the Outlet.

Dip Cycles

If you wish to include a dip cycle after the sample injection step, use a solution of ampholytes as the Inlet and Waste as the Outlet. For a dip cycle after injection of the standards, select a solution of the anolyte as the Inlet and Waste as the Outlet. Before creating a Method, make sure to enter the internal standards and the uniquely-labeled buffers (Ampholyte-Dip, anolyte-Dip, etc.) in the reagent database and your Configuration.

Focusing and Mobilization

Focusing and Mobilization conditions are defined from the Focusing/Mobilization Parameters Table at the bottom of the Setup screen. The Define Method screen for the IEF method template comes up with default parameters in place: for both focusing and mobilization steps, IEFAnalyte has been assigned as the Inlet reagent, whereas IEF Catholyte and IEF Mobilizer have been selected as the Outlet focusing and mobilization reagents, respectively. These may be changed by double-clicking the reagent, then selecting the desired reagent from the reagent database window. You may terminate the focusing process by time or by current minimum (default values are 240 seconds or 0.3 μA at 15 kV).

The recommended current limit during focusing and mobilization is 100 μA , well above the current levels observed when using Bio-Rad CE-IEF reagents and recommended run conditions. To terminate the focusing process by a current minimum, double-click **Time (sec)**, and the next screen appears. See Figure 5-61.

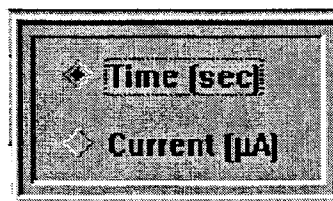


Figure 5-61 Time/Current Selection Box

Make your selection by clicking your choice. The default values of 240 seconds or 0.3 μ A can be changed by entering the desired values in the **END** value column.

The example shown in Figure 5-62 shows an IEF method with three of its Post Inject prep cycles defined: a low pressure dip cycle (0 psi * sec) to rinse the exterior of the capillary, a low-pressure multi inject step (20 psi * sec) to load the standards, and another low-pressure dip cycle (0 psi * sec) to rinse the exterior of the electrode and capillary after injection of the standards.

The 'Define Method' dialog box is shown with the following fields and tables:

Method ID: **HEMOGLOBINS** OK Cancel

Description: **IEF method** Polarity: $\leftarrow + \rightarrow -$ $\rightarrow - \leftarrow +$

Inject Time: **20** sec Run Time: **30.00** min Capillary Temperature: **27** °C

Detector: **Single Wavelength Mode, 288 nm**

Timed Events

Preparation Cycle Parameters: 3 cycle[s] defined

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Post-inj	IEFANOLYTE	Waste	Low	0	psi*sec
2	Post-inj	INTERNALSTD	Waste	Low	20	psi*sec
3	Post-inj	IEFANOLYTE	Waste	High	0	sec

Buttons: Add Insert Delete Delete All

Step	Inlet	Outlet	Terminate Focus by	End Value	Voltage (kV)	Limit (pA)
Focus	IEFANOLYTE	IEFCATHOLYTE	Time(sec)	240	15.0	100.0
Mobilize	IEFANOLYTE	IEFMobilizer			15.0	100.0

Figure 5-62 IEF Method With Three Post Inject Prep Cycles Defined

Saving the Method

When you finish defining all run parameters, click **OK** to save the new IEF method before returning to the main window. If you click **Cancel**, you will exit the Define Method screen without saving the new data that you have entered.

The 'Browse Methods' dialog box displays a list of methods with the following columns: Method ID, Type, Detector, and Method Description.

Method ID	Type	Detector	Method Description
CZE_TEMPLATE	CZE	A	peptide method
CZE_TMPL_LIF	CZE	F	peptide method
EQUILIBRATE	CZE	A	Equilibration Method
FINAL	CZE	A	Terminates Autosequence
IEF_TEMPLATE	IEF	A	IEF method
IEF_TMPL_LIF	IEF	F	IEF method
PCR	CZE	A	PCR Product method
SDS_PROTEIN	CZE	A	SDS Protein method
SHUTDOWN	CZE	A	terminates autosequence
SHUTDOWN_LIF	CZE	F	terminates autosequence
VERIFY	CZE	A	Verification Method

Buttons: Define Edit Delete Delete All Exit

Figure 5-63 Browse Methods Screen

Editing an IEF Method

Just as with CZE, your new IEF method can be edited by clicking the **Edit** button in the Browse Methods screen shown in Figure 5-63. The IEF template and IEF methods linked to automation sequences cannot be edited.

To leave the Browse Methods screen and return to the main screen, click **Exit**.



CAUTION! Although your computer can be used for multitasking during most runs, you should avoid doing so during IEF if you are monitoring the current closely.

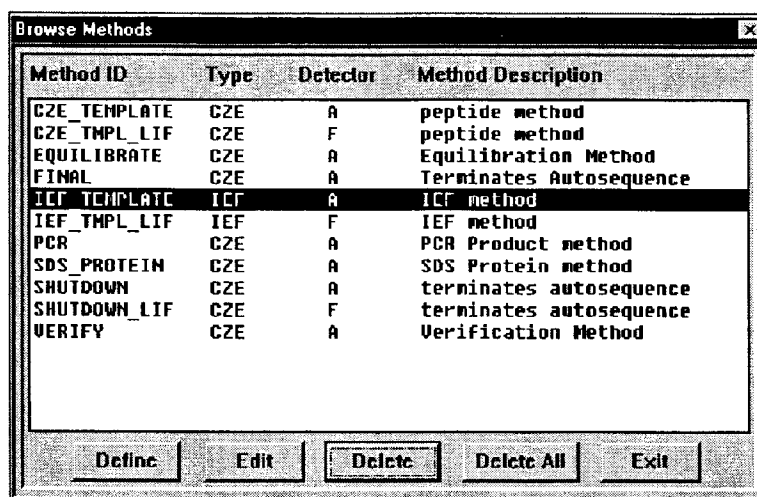


Figure 5-64 Browse Methods Screen



Print Requests Button

If you wish to make a hard copy of the new method, you can do so from the Print Requests button on the toolbar (this button becomes inactive during the run).

Defining Other Methods

The CZE template may also be used for defining other methods, including Micellar Electrokinetic Chromatography (MEKC), Capillary Ion Analysis (CIA), Capillary Gel Electrophoresis (CGE), and Chiral Analysis.

Single Run Group



Single Run Group Button

Overview

Single Run Group methods can be accessed from the **Commands** menu or by clicking the **SRG** button.

The Single Run Group capability of BioFocus software allows you to run a single CZE or IEF method with one or more samples. This feature is designed for single analysis or for hands-on method development where full automation is unnecessary or undesirable. For more complex automated programming, you will need to work with the Main Screen buttons. Single Run Groups have the following limitations:

1. Only a single CZE (or IEF) method may be programmed. Once programmed, method parameters will be saved only when the **Start** or **Save and Exit** button is pressed. Each sample vial can only be sampled once, thus replicate runs can only be performed by pipetting aliquots of a single sample into multiple vials.
2. Prep, inlet run buffer, sample, outlet run buffer, waste, and fraction positions are fixed and appear on the Setup for Single Run Group Automation Sequence screens.
3. Only one waste position is available.
4. No priority scheduling is available.
5. No cartridge use history is recorded.
6. Only up to a total of six prep cycles can be defined.

The Single Run Group CZE and IEF Methods are described in detail in the following paragraphs.

CZE Method

The first screen, Single Run Group, provides a choice of CZE or IEF. Select one of these and then click OK. Refer to Figure 5-65.

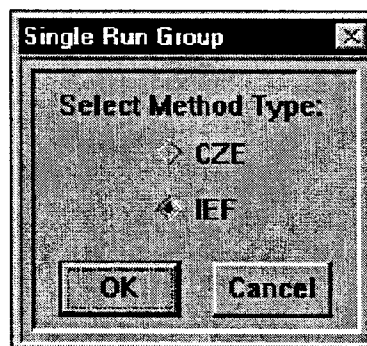


Figure 5-65 Single Run Group Box

NOTE: Vial positions for the CZE Single Run Group method are fixed and are indicated on the Setup screen. Inlet Carousel Vial Positions 1 through 6 have been reserved for prep cycle reagents, position 7 is for the inlet run buffer and positions 8 – 32 are for samples. The Outlet Carousel Position 1 is for the waste vial, position 2 is for the outlet run buffer, and positions 3 – 32 can be used for fractions.

Programming a CZE Single Run Group

If you select CZE, a Setup for CZE Single Run Group Automation Sequence screen appears. See Figure 131. On this screen, you must define the total Number of Samples (up to 25), Run Voltage (or current), Current Limit (or voltage limit), Polarity, Capillary Temperature, Run Time, Prep Cycles (up to six), and optional Fractions (up to 30).

Setup for CZE Single Run Group Automation Sequence

Number of Samples in inlet carousel position(s): Start Cancel

Fractions fraction vial(s) selected. Save and Exit

Run Buffer Positions: Inlet at 7, Outlet at 2

Run Parameters:

☒ Voltage Mode Run Voltage kV Polarity: ☐ + ☐ - ☐ + ☐ -
☐ Current Mode Current Limit uA Capillary Temperature °C
 Run Time minute(s)

Preparation Cycle Parameters: 1 cycle(s) defined

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Pre-inj	Position 1	Position 1	High	60	sec

Add Delete Delete All

☐ Injection Pressure: 20 (psi * sec)
☐ Detector
☐ Timed Events

Figure 5-66 Setup for CZE Single Run Group Automation Sequence

Fractions Collection

If you wish to collect Fractions during a run group, click the **Fractions** button at the upper right corner of the Define Run Groups for Automation Sequence screen and a Fractions table with all the fraction vials defined in the SRG configuration appears.

Double-click on an Outlet Carousel Vial and it appears under Fraction Name. Enter a Start Time and End Time for the fraction; click the **Save** button. Repeat for additional fractions. When you have entered all fractions in the table that you wish to activate, click **Save Table and Exit** to return to the Setup screen for CZE Single Run Group. Clicking the **Delete All Times** button clears all the start and end times you entered previously.

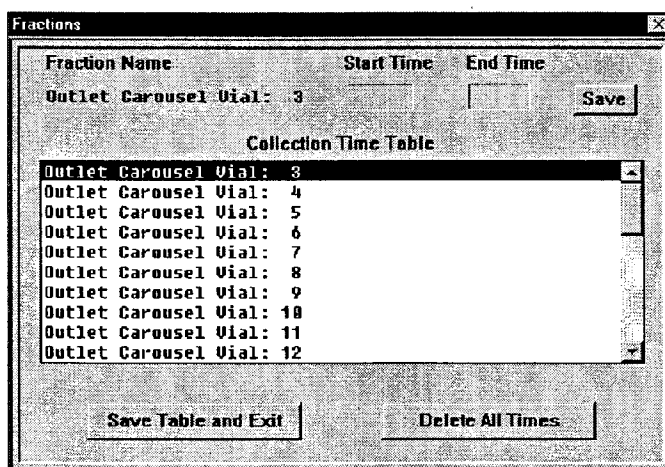


Figure 5-67 Fractions Screen

Entering Start and End Times

The Fractions screen shown in Figure 5-68 displays the name of the first fraction under the Fraction Name at the top left corner of the screen. Based on data from earlier trial runs, enter the peak Start Time and End Time for this fraction, then click **Save**. For End Time, up to 99.99 minutes can be programmed. The start and end times for the first fraction now appear in the Collection Time Table shown in Figure 5-68.

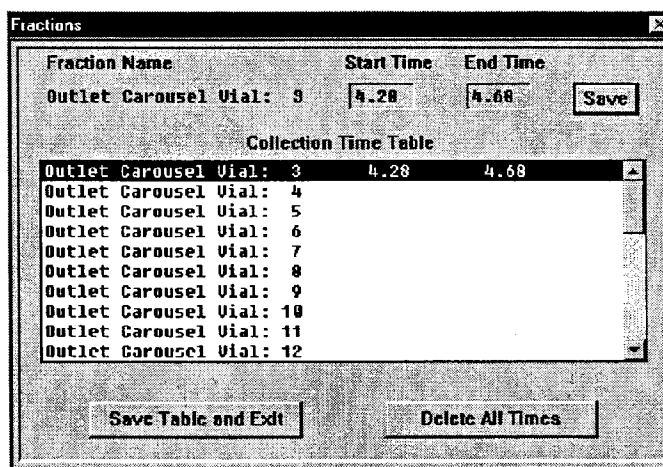


Figure 5-68 Fractions Screen—Start and End Times Entered

Double-click the next fraction that you wish to activate in the Collection Time Table.

It now appears under Fraction Name. Enter Start and End times and click **Save**. See Figure 5-69.

The screenshot shows a window titled "Fractions" with a table for entering fraction collection data. The table has three columns: "Fraction Name", "Start Time", and "End Time". The first row is "Outlet Carousel Vial: 4" with "Start Time" 5.34 and "End Time" 5.74. A "Save" button is to the right of the first row. Below this is a "Collection Time Table" with a scrollable list of vials from 3 to 12. The first two rows of this table are filled with data: "Outlet Carousel Vial: 3" with times 4.28 and 4.68, and "Outlet Carousel Vial: 4" with times 5.34 and 5.74. At the bottom are two buttons: "Save Table and Exit" and "Delete All Times".

Fraction Name	Start Time	End Time
Outlet Carousel Vial: 4	5.34	5.74

Outlet Carousel Vial	Start Time	End Time
Outlet Carousel Vial: 3	4.28	4.68
Outlet Carousel Vial: 4	5.34	5.74
Outlet Carousel Vial: 5		
Outlet Carousel Vial: 6		
Outlet Carousel Vial: 7		
Outlet Carousel Vial: 8		
Outlet Carousel Vial: 9		
Outlet Carousel Vial: 10		
Outlet Carousel Vial: 11		
Outlet Carousel Vial: 12		

Figure 5-69 Fractions Screen—Second Fraction Entered

Successful fraction collection requires that run conditions be identical to those of your trial runs on which peak times are based.

During fraction collection, the system automatically calculates the transit time for each peak to migrate from the detector window to the capillary outlet. It will turn the high voltage on and off at appropriate times and rotate the outlet carousel to each fraction position. The system also keeps track and compensates for the voltage-off and carousel rotation times. Consequently, the real time electropherograms may have spikes and very broad peaks. Do not attempt multitasking or integration while collecting fractions. Make sure that the Auto Integration function is deselected.

When collecting fractions with a CZE method, the software calculation of detector-to-outlet transit time employs the capillary length of the cartridge. See "Entering Capillary Length" on page 5-75.

When collecting fractions with an IEF method, the detector-to-outlet transit time cannot be accurately predicted due to variations in focusing time and mobilization rates. Therefore no software compensation for transit time is made. These must be determined experimentally for each IEF method. A preferable approach is collection of back-to-back fractions of equal times with post-run analysis of collected fractions. Do not attempt fraction collection during IEF focusing.

The amount of material collected in a fraction will depend on the concentration of the component in the sample, the injection parameters, and the capillary dimensions. For collection of useful amounts of material, capillaries of 50 μm ID or greater are recommended. Since only nanoliter volumes are typically collected in a single run, it may be necessary to collect fractions from replicate runs in the same vial to obtain sufficient material for analysis. You need to have enough (15 μl minimum, preferably 50 μl) buffer in the collection vial to complete the circuit and also compensate for possible evaporation during multiple runs.

When you have completed all fraction selections, click **Save Table and Exit**.

Defining Preparation Cycle Parameters

Next, you need to define the Preparation Cycle Parameters for washing the capillary, filling the capillary with run buffer, rinsing the outer surface of the capillary and electrode before or after sample injection, or for scheduling multiple injections (Multi Inject).

Defining Injection Type

You can define the injection type and associated parameters by clicking the **Injection** button at the lower left corner of the Setup screen.

Defining Detector Parameters

To define detection parameters, simply click the **Detector** button at the lower left corner of the Setup screen.



Print Requests Button

Making Hard Copies

If you wish to make hard copies of the CZE Single Run Group Configuration form and/or Method, you can do so by clicking the Print Requests button on the Main Screen toolbar.

Saving Single Run Groups

It is not possible to save more than one CZE (or IEF) Single Run Group program. If you wish to save more than one method, you must program through the Main Screen buttons. If you wish to obtain the original default values after using the Single Run Group for multiple purposes, consult the values of the CZE or IEF templates from the Browse Methods screen.

Starting a Single Run Group

Be sure there is a cartridge in place, coolant in the reservoir, and that the gas tank valve is open. When you are ready to begin, click **Start**.

Entering Capillary Length

If you have programmed fraction collection in your CZE method, a window appears (see Figure 5-71.), requesting you to enter a cartridge length. Enter the capillary length and click **OK** to continue the run. The computer uses this information to calculate fraction transit times.

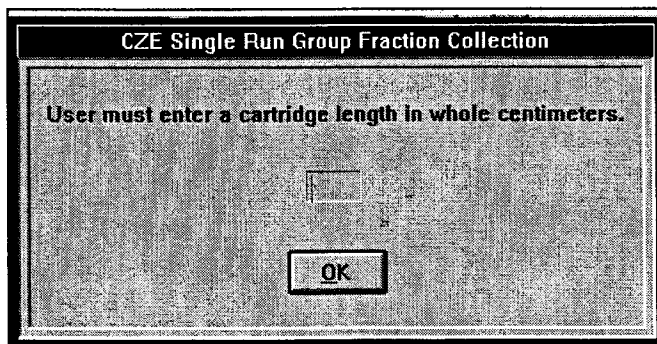


Figure 5-70 Single Run Group Fraction Collection-Enter Cartridge Length

If you enter a number less than 12 cm, a warning message appears. See Figure 5-71.

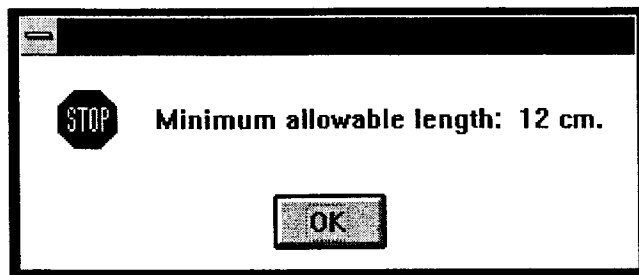


Figure 5-71 Warning Message For Entry Below Minimum Allowable Length

Correct your entry in the Single Run Group Fraction Collection screen and click **OK** to continue.

IEF Method

You can easily program an IEF method if you click the **IEF** button and then on **OK** in the Single Run Group screen shown in Figure 5-72.

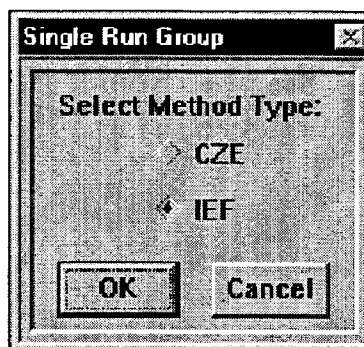


Figure 5-72 Single Run Group Selection Box-IEF

NOTE: Vial positions for the IEF Single Run Group method are fixed and appear on this screen.

The next screen which appears is the Setup for IEF Single Run Group Automation Sequence. See Figure 5-73.

Vial Positions

Inlet Carousel Vial Positions 1 through 6 are for prep cycle reagents. Position 7 must contain the inlet run buffer (anolyte) for the focusing and mobilization steps and positions 8-32 are reserved for samples.

Outlet Carousel Vial Position 1 is for the waste vial. Position 2 must have the outlet run buffer (catholyte) for the focusing step, position 3 is for the outlet run buffer (mobilizer) for the mobilization step and positions 4 – 32 can be used for fractions.

Setup for IEF Single Run Group Automation Sequence

Number of Samples: 1 in inlet carousel position(s): 8 Start Cancel

Fractions: 0 fraction vial(s) selected. Save and Exit

Run Buffer Positions: Inlet at 7, Outlet at 2

Run Parameters:

- Voltage Mode: Run Voltage: 10.0 kV Polarity: $\leftarrow + \rightarrow -$ $\rightarrow - \leftarrow +$
- Current Mode: Current Limit: 100.0 μ A Capillary Temperature: 20 $^{\circ}$ C
- Run Time: 0.00 minute(s)

Preparation Cycle Parameters: 1 cycle(s) defined

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Pre-inj	Position 1	Position 1	High	60	sec

Add Delete Delete All

Injection: Pressure, 20 (psi * sec)

Detector

Timed Events

Figure 5-73 Setup For IEF Single Run Group Automation Sequence

Collecting Fractions

You must define the total Number of Samples (up to 25), optional **Fractions** (up to 29), Inject time, Run Time, Polarity, Capillary Temperature and Detection, Prep Cycle (up to six) and Focusing/Mobilization Parameters.

Fractions may be collected during an IEF separation by programming collection times in the fraction collection time table as described for the CZE Single Run Group method. However, the software does not compensate for transit time from the detection window to the capillary outlet in IEF. This must be determined experimentally for each mobilization method. A preferable approach is collection of back-to-back fractions of equal times and post run analysis of collected fractions. *Do not attempt fraction collection during focusing.*

Injection Time

For a 50 μm x 24 cm capillary, the recommended injection time is 20 seconds at 100 psi. Run time is combined for both focusing and mobilization. The suggested detection wavelength is 280 nm. Polarity is set positive to negative.

NOTE: The Bio-Rad supplied IEF template comes with no prep cycles defined.

Prep Cycles

Prep cycles can be defined from the Preparation Cycle Parameters Table.

Focusing and Mobilization

Focusing and Mobilization conditions are defined from the Focusing/ Mobilization Parameters Table at the bottom of the Setup screen. You may terminate the focusing process with a time limit or a current minimum (default values are 240 seconds at 15 kV or 0.3 μA). The recommended current limit during focusing and mobilization is 100 μA . If you wish to terminate the focusing process by a current minimum, double-click **Time (sec)**, and the next screen appears. Refer to Figure 5-74.

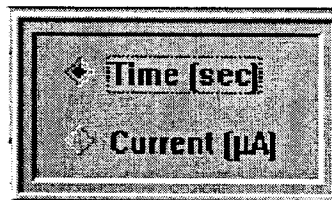


Figure 5-74 Terminate Focusing Process Options Box

Make your selection by clicking your choice. Your selection will now automatically appear in the Terminate Focus By column, and you can enter the desired value in the End Value column.

When you finish defining all run parameters, click **Save and Exit** to save the new IEF Single Run Group Automation Sequence before re-

turning to the main window. Clicking **Start** will also save the new parameters. If you click **Cancel**, the data you have entered will not be saved.



Print Requests Button

Hard Copies

If you wish to make hard copies of the IEF Single Run Group Configuration and/or Method, click the **Print Requests** button on the toolbar.

LIF Single Run Groups

Defining a single run group using LIF detection is identical to defining a single run group using absorbance detection, except for detector setup. If you click the SRG button in LIF mode, the screen shown in Figure 5-75 appears. Note that the detector indication lists laser wavelength and PMT gain. To turn lasers on and set PMT gain, click **Detector**. The screen shown in Figure 5-76 appears.

Setting Parameters

To set parameters for a single run, click any of the buttons and enter the desired values. Ensure that you turn on internal lasers when making settings in the LIF Parameters dialog box. If you did not turn on at least one of the lasers or you set both PMT gains to zero in the LIF Parameters screen, an error message (see Figure 5-54) appears to remind you to make the correct settings.

Setup for CZE Single Run Group Automation Sequence

Number of Samples: 1 in Inlet carousel position[s]: 8

Fractions: 0 fraction vial[s] selected.

Run Buffer Positions: Inlet at 7, Outlet at 2

Run Parameters:

- Voltage Mode: Run Voltage: 10.0 kV
- Current Mode: Current Limit: 100.0 uA
- Polarity: + + + - - - +
- Capillary Temperature: 20 °C
- Run Time: 5.00 minute[s]

Preparation Cycle Parameters: 2 cycle[s] defined

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Pre-inj	Position 1	Position 1	High	60	sec
2	Post-inj	Position 2	Position 1	High	30	sec

Buttons: Add, Delete, Delete All

Injection: Pressure: 20 (psi * sec)

Detector: Laser 488 nm Laser Ext PMT1 10 X Gain PMT2 10 X Gain

Timed Events

Figure 5-75 LIF Single Run Group Screen

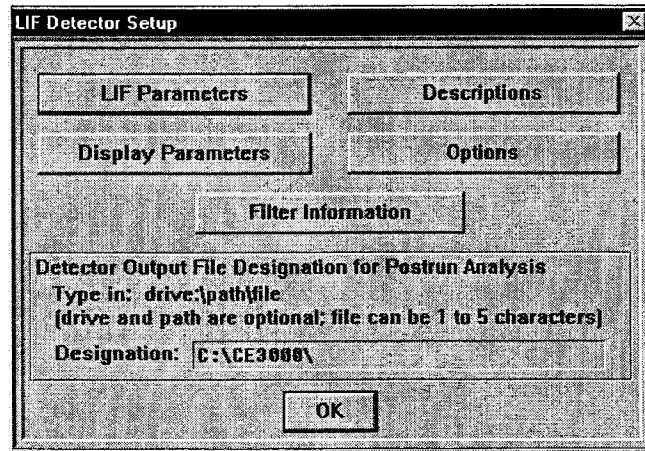


Figure 5-76 LIF Detector Setup Screen